

Global Poverty & Inequality Trends: A WDI Analysis

This project applies the end-to-end data analytics workflow to examine global poverty and inequality trends using the World Bank's World Development Indicators (WDI) dataset. It aims to identify patterns in poverty headcount rates, income inequality (via the Gini index and income share distribution), and their relationship to economic and social indicators such as GDP per capita, unemployment, and literacy, in order to surface actionable insights for policymakers and development stakeholders.

Dataset Description

The dataset used in this project is the World Bank's World Development Indicators (WDI), the World Bank's flagship collection of global development statistics covering over 200 countries and territories. The full dataset (WDICSV.csv) contains 396,970 records spanning 1,400+ indicators from 1960 to 2025. For this analysis, the dataset was filtered to 10 indicators directly relevant to poverty and inequality: poverty headcount ratio (at \$3.00/day and at national poverty lines), poverty gap, Gini index, income share held by the highest and lowest 10% of earners, GDP per capita, unemployment rate, adult literacy rate, and total population. Data was retained for both individual countries and World Bank aggregate groupings (e.g., regions, income groups), flagged separately to prevent double-counting in country-level comparisons.

Data Cleaning Process

Data cleaning and transformation were performed in Power BI's Power Query Editor. The following steps were applied:

1. Filtered the raw dataset (396,970 rows across 1,400+ indicators) down to the 10 indicators relevant to this analysis.
2. Reshaped the data from wide format (one column per year) to long format (one row per Country, Indicator, and Year) using an unpivot operation, enabling proper time-series analysis.

3. Removed rows with missing values, since WDI data has significant gaps — particularly for poverty and inequality indicators, which are collected through periodic household surveys rather than annually.
4. Removed duplicate rows.
5. Standardized data types (Year as whole number, Value as decimal).
6. Added a calculated "Row Type" column to distinguish individual countries from World Bank aggregate groupings (e.g., World, regions, income groups), preventing double-counting in comparisons.
7. Pivoted the cleaned long-format data back into a wide format (one row per Country and Year, with each indicator as its own column) to enable calculated fields.
8. Added a calculated "Income Share Ratio" column (income share of the highest 10% divided by the lowest 10%) as a derived inequality measure.

A key data limitation identified during cleaning: poverty and inequality indicators (Gini index, poverty headcount ratios) have far sparser coverage than indicators like population or GDP, since they depend on household survey cycles that are not conducted annually in every country.

Analysis Methodology

The analytical workflow followed a standard data analytics pipeline: data acquisition, cleaning and transformation, exploratory analysis, and visualization. All cleaning and transformation were performed in Power BI's Power Query Editor, as described above. Analysis was structured around seven core angles, each mapped to a specific dashboard visual:

1. KPI snapshot cards: summarizing average Gini index, average poverty rate, number of countries with data, and the latest year covered, all calculated using DAX measures filtered to the most recent available year.
2. Trend analysis: a line chart tracking the average Gini index across countries over time (1960–2025).

3. Relationship analysis: two scatter plots examining (a) GDP per capita vs. poverty rate, and (b) unemployment rate vs. poverty rate, to explore correlations between economic conditions and poverty.
4. Regional comparison: a bar chart comparing average poverty rates across World Bank regional and income-group aggregates for the most recent year with available data (2024).
5. Inequality ranking: a bar chart ranking the top 10 countries by income share ratio (income share of the top 10% divided by the bottom 10%), a derived measure of income concentration.
6. Change-over-time analysis: a table comparing each country's earliest and latest available poverty headcount rate, calculated via DAX measures, to identify the most improved and most worsened countries.

Key Findings

1. **Global inequality has risen over time.** The average Gini index across countries with available data rose from roughly 30–35 in the 1960s–1980s to over 40 in recent years, indicating a long-term trend toward greater income inequality globally.
2. **Extreme poverty has declined substantially on average,** but coverage and severity vary widely. The average poverty headcount rate (at \$3.00/day) fell from an earliest recorded average of 28.97% to a considerably lower figure in the most recent data, though this masks sharp regional and country-level divergence (see below).
3. **Poverty remains heavily concentrated in specific regions.** Sub-Saharan Africa and the IDA (International Development Association) country grouping recorded the highest average poverty rates in 2024, substantially above other regions and income groups.
4. **Conflict-affected countries show the sharpest recent increases in poverty,** not economic factors alone. The countries with the largest poverty rate increases from their earliest to latest recorded values were South Sudan (+31.6 percentage points), Yemen (+19.2), Syria (+16.8), Kenya (+13.5), and Zimbabwe (+13.5), all countries that experienced significant conflict, political instability, or economic crisis during their measured periods.

5. **Some countries show major poverty improvement.** Armenia (–20.8 points), Algeria (–12.3), and Albania (–2.7) recorded the largest declines in poverty headcount rate between their earliest and latest available data points.
6. **Income inequality is highest in several Latin American countries.** Colombia, Panama, Brazil, Honduras, and Ecuador rank as the top countries by income share ratio (the income share held by the top 10% relative to the bottom 10%), consistent with well-documented patterns of income concentration in the region.
7. **GDP per capita and unemployment show a general inverse relationship with poverty,** though with notable outliers, several countries combine relatively low GDP per capita with low measured poverty rates, and vice versa, suggesting other structural factors (social protection systems, informal economies, data collection quality) also play a significant role.

Insights

1. **Poverty and inequality do not move in parallel:** Poverty rates have generally declined while the Gini index has risen, meaning fewer people are in extreme poverty, but income is being shared less evenly among those above the poverty line. Reducing poverty and reducing inequality require different policy levers.
2. **Fragility, not just low income, drives poverty spikes:** The largest recent increases in poverty are concentrated in conflict- and crisis-affected countries rather than simply low-GDP countries, suggesting that stability and governance may matter as much as economic growth for poverty outcomes.
3. **High-inequality countries are not necessarily the poorest:** Several countries with high income share ratios (e.g., Brazil, the United States) are middle- or high-income economies, showing that inequality is a distinct problem from poverty, capable of persisting even amid overall prosperity.
4. **Data availability itself is an equity issue:** Poverty and inequality indicators have far sparser coverage than population or GDP data, meaning the countries with the weakest statistical infrastructure, often the poorest, are also the hardest to monitor and hold accountable through data.

Recommendations

1. Prioritize poverty-reduction interventions in Sub-Saharan Africa and IDA-eligible countries, where poverty rates remain highest and improvements have been slowest, rather than applying uniform global strategies.
2. Treat conflict and fragility as a poverty risk factor requiring dedicated response mechanisms, the sharp poverty increases in South Sudan, Yemen, and Syria suggest that humanitarian and stabilization support should be paired with, not separate from, economic development programs in crisis-affected states.
3. Address inequality independently of poverty reduction, particularly in middle- and high-income countries with high income concentration (e.g., several Latin American nations). Progressive taxation, wage policy, and social protection expansion are relevant levers even where extreme poverty is already low.
4. Invest in statistical infrastructure for countries with the weakest data coverage, since poverty and inequality cannot be effectively targeted or monitored without regular, reliable household survey data.
5. Track both poverty and inequality metrics together, not poverty alone, in national and international development reporting, since this analysis shows that poverty reduction is not sufficient evidence of broadly shared economic progress.

Conclusion

This analysis of World Bank WDI data on poverty and inequality reveals a complex global picture: extreme poverty has declined on average, yet income inequality has risen, and poverty remains acutely concentrated in specific regions and crisis-affected countries. The findings underscore that poverty and inequality are related but distinct challenges, each requiring targeted policy responses. Data limitations, particularly sparse coverage in the countries most affected, also highlight the need for continued investment in development statistics. The accompanying interactive Power BI dashboard allows stakeholders to explore these patterns further by country, region, and year.

